How-To Library

Informatica

Using SSL to Connect to a Greenplum Database in PowerExchange for GreenPlum

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Abstract

This article describes how to configure secure communication between PowerExchange for Greenplum and the Greenplum server by using the Secure Sockets Layer (SSL) protocol. This article also describes the SSL configurations required for the DataDirect ODBC driver to connect to the Greenplum database from Informatica.

Supported Versions

- PowerExchange for Greenplum 10.2 or later

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Connecting to a Greenplum Database Using SSL Overview

When you create a Greenplum connection, the gpload utility uses the configured attributes to connect to the Greenplum database. You can configure secure communication between the gpload utility and the Greenplum server by using the Secure Sockets Layer (SSL) protocol. SSL is a protocol that ensures secure data transfer between a client and a server.

To enable PowerExchange for Greenplum for secure communication between the gpload utility and the Greenplum server, you must select the Enable SSL option in the Greenplum connection. Generate the server and client certificates. In the Greenplum connection, define the path where you stored the SSL certificates for the Greenplum server.

You can also configure the DataDirect ODBC driver on Windows or UNIX to connect to a Greenplum Database from Informatica using SSL.

Generate the Server Self-Signed Certificates

You must generate the `root.crt`, `server.crt`, and `server.key` self-signed certificates on the Informatica node that you configured to run gpload.

1. Generate the `root.crt`, `server.crt`, and `server.key` on the Informatica node configured for gpload.

   For example, run the following commands to generate the certificates:

   ```
   openssl genrsa -des3 -out server.key 1024
   openssl rsa -in server.key -out server.key
   chmod 400 server.key
   openssl req -new -key server.key -days 3650 -out server.crt -x509 -subj
   '/C=IN/ST=Karnataka/L=Bangalore/O=Informatica.com/CN=INVR28PERS47/
   emailAddress=info@informatica.com'
   cp server.crt root.crt
   ```

2. Copy the generated files to a directory that gpload can access.
3. Set the gpload SSL path to point to the directory that contains the keys.
4. Ensure that the CN value is the same as the host name of the Informatica node.

Generate the Client Self-Signed Certificates

You must generate the `client.key`, `client.crt`, and `client.csr` self-signed certificates on the Informatica node that you configured to run gpload.

1. Generate the `client.key`, `client.crt`, and `client.csr` certificates.

   For example, run the following commands to generate the certificates:

   ```
   openssl genrsa -des3 -out client.key 1024
   openssl rsa -in client.key -out client.key
   openssl req -new -key client.key -out client.csr -subj
   '/C=IN/ST=Karnataka/L=Bangalore/O=Informatica.com/CN=adpqa'
   openssl x509 -req -in client.csr -CA root.crt -CAkey server.key -out
   client.crt -Cacreateserial
   ```

2. Copy the generated files to the segment directory in the Greenplum server.

   For example, `/data/disk1/primary/gpseg0`

3. Copy the keys to the following directory:

   `For example, /data/disk1/primary/gpseg0/gpfdists`

Configure PowerExchange for Greenplum to connect to a Greenplum Database using SSL

Use a Greenplum connection to connect to a Greenplum database from Informatica. The Greenplum connection is a relational type connection. You can create and manage a Greenplum connection in the Administrator tool or the Developer tool.

PowerExchange for Greenplum uses the host name, port number, and database name to create a control file to provide load specifications to the Greenplum gpload bulk loading utility. Enable the SSL option and provide the path to the SSL certificates for the Greenplum server to establish secure communication to the Greenplum server.

1. Click **Window** > **Preferences** in the Developer Tool.
2. Select **Informatica** > **Connections**.
3. Expand the domain in the Available Connections.
4. Select **Databases** > **Greenplum Loaders**, and click **Add**.
5. Enter a connection name and an ID for the connection.
6. Select the domain on which you want to create the connection.
7. Select a Greenplum Loader connection type, and click **Next**.
8. Configure the connection properties for metadata access and data access.
9. Select the **Enable SSL** option to establish secure communication between the gpload utility and the Greenplum server over SSL.
10. Specify the path where the SSL certificates for the Greenplum server are stored.
11. Click **Test Connection** to verify the connection to Greenplum.
12. Click **Finish**.

Configure DataDirect ODBC on Windows to Connect to a Greenplum Database Using SSL

You can securely connect to the Greenplum database from Informatica installed on Windows with the DataDirect ODBC driver.

1. Configure the DataDirect ODBC driver on the machine hosting the Informatica Services.
2. Enter the configuration for the Data Source Name (DSN) by setting the following options on the **Security** tab of the ODBC Greenplum Driver Setup page:
   - EncryptionMethod = 1 - SSL
   - (Optional) TrustStore = <location_of_truststore_file>
   - (Optional) TrustStorePassword = <password used to access the truststore file>
   - ValidateServerCertificate = 0 or 1

Configure DataDirect ODBC on UNIX to Connect to a Greenplum Database Using SSL

You can securely connect to the Greenplum database from Informatica installed on Windows with the DataDirect ODBC driver.

1. Install and configure the DataDirect ODBC drivers on the machine hosting the Informatica services.
2. Perform one of the following steps:
   - Edit the existing `odbc.ini` file in the `$ODBCHOME` directory.
   - Copy the `odbc.ini` file to the root directory and edit it.
     For example, `cp $ODBCHOME/odbc.ini $HOME/.odbc.ini`
3. Add an entry for the ODBC data source under the ODBC Data Sources section, and configure the data source.

   The following syntax shows an example of the configurable options for the ODBC data source:

```
[gp43ssl_adpqa_db]
Driver=/opt/Informatica//ODBC7.1/lib/DWgplm27.so
Description=DataDirect 7.1 Greenplum Wire Protocol
AlternateServers=
ApplicationUsingThreads=1
ConnectionReset=0
ConnectionRetryCount=0
ConnectionRetryDelay=3
Database=adpqa_db
ConnectionReset=0
ConnectionRetryCount=0
ConnectionRetryDelay=3
Database=adpqa_db
EnableDescribeParam=1
EnableKeysetCursors=0
EncryptionMethod=1
ExtendedColumnMetaData=0
FailoverGranularity=0
FailoverMode=0
FailoverPreconnect=0
FetchRefCursors=1
FetchTSWTZasTimestamp=0
FetchTWFSasTime=0
HostName=gp43ssl.informatica.com
HostNameInCertificate=
IANAAppCodePage=4
InitializationString=
KeyPassword=
```

```
KeysetCursorOptions=0
KeyStore=/thirdparty/greenplum/keystore/keystore.pkcs12
KeyStorePassword=Informatica
LoadBalanceTimeout=0
LoadBalancing=0
LoginTimeout=15
LogonID=adpqa
MaxCharSize=
MaxLongVarcharSize=
MaxPoolSize=100
MaxVarcharSize=
MinPoolSize=0
Password=adpqa
Pooling=0
PortNumber=5432
QueryTimeout=0
ReportCodepageConversionErrors=0
TransactionErrorBehavior=1
TrustStore=
TrustStorePassword=
ValidateServerCertificate=0
XMLDescribeType=-10
```

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